Market integration in a transitional hunter-gatherer population benefits men, not women

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Abstract
Background and Objectives: The transition from hunting-gathering to domestication and market economics creates disparity in wealth and health. Comparing past and contemporary transitions is challenging due to differences in data. We ask whether methodological differences have failed to capture sex-based health differences in past transitional societies.
Methodology: We collected data in living hunter-gatherers for a common marker of poor health in past archaeological populations, missing enamel that indicates an interruption in growth during childhood due to malnutrition or disease. We analyzed data using t-tests with sex as a factor. We then analyzed data using additional information available in living populations, year of birth, via a time series regression.
Results: Without year of birth included in analysis, no sex-based differences are found, which makes sense given that hunter-gatherers establish minimal social hierarchies beyond a sex-based division of labor. When including year of birth, we observe a clear difference between men and women. Starting about 40 years ago, younger generations of Hadza men experienced less childhood stress than women and older generations.
Conclusions and Implications: Economic and subsistence transitions often cause negative impacts to health. Sometimes transitions differentially benefit one sex (usually male) or disadvantage the other (usually female). The transition to a market economy in bush-living Hadza has benefitted men's health, but not women's health. The broader implications are that the sex-based division of labor in hunting-gatherer societies is exacerbated by external forces favoring males over females.

Lay Summary
The transition from hunting-gathering to market economies creates wealth and health disparity, often gendered. We applied methods for studying archaeological populations in a living human population and found men's health improves in response to market integration and women's does not.

BACKGROUND AND OBJECTIVES

The recent origins of societal inequality are often linked with sedentism and agriculture (1, 2) because, aside from a sex-based division of labor (3), hunter-gatherers retain high levels of individual autonomy and are egalitarian. While social status differences exist in egalitarian societies, they are more subtle than those in hierarchical systems (1, 4). The hunter-gatherer sex-based division of labor (3), however, initializes role specialization very early in life and may be the first instance of tacitly permissible social inequality (5, 6). We apply methods for studying health and inequality in archaeological populations to living hunter-gatherers to understand if subtle gender-based inequalities affect health in a similar manner as observed in stratified societies.

Bioarchaeological research across the globe indicates that past populations suffered from higher rates of lethal and chronic infectious diseases, nutritional deficiencies, and biomechanical problems introduced via labor when transitioning from hunting-gathering to domestication and agricultural economies (7). Contemporary societies in transition often have health declines similar to those documented in the past, but health problems observed in the shift to an industrialized market economy are more pronounced than what has been observed in the past when shifting from foraging to small-scale food producing economies (8-10). Some transitions to market-integrated lifestyles have been relatively beneficial to some populations, such as the indigenous Venezuelan horticulturalists, the Pumé, who experienced improvements in some health measures due to greater accessibility of agricultural and market foods (e.g., fewer and less severe incidents of nutritional stress at weaning) (11). Even a simple shift from foraging to sedentism has resulted in a worsening of health and reduced mortality in Agta in the Philippines (12) and the same occurs in the shift to urban environments in Turkana pastoralists—in the latter, the advantages of status within the community disappears in urban settings (13). Successful transitions, however, are rare and most hunter-gatherers find themselves at the bottom of a new social and economic hierarchy, possessing little wealth (14) and experiencing restricted land availability. In contemporary societies, sex-based health disparity is common (15-17) and that may stem from the initial sex-based division of labor, as suggested above.

Why do we see a shift from little to no sex-based health disparity to sex-based health disparity? On a theoretical level, there are critical differences in the context surrounding the motivation for transition between past and contemporary peoples. Past hunter-gather subsistence transitions were shaped by ecologically salient forces while contemporary ones are typically shaped by external socio-political forces (9). Unlike the transition to agricultural subsistence that occurred over 10,000 years ago and the subsequent transition to the use of currency and a market economy hundreds of years ago, the transition of extant hunter-gatherers into either of these systems (typically a blend) is rapid and externally motivated (18). On a methodological level, the shift to increased evidence for sex-based health differences might be an artifact of the types of data available in past versus contemporary societies. Rarely do archaeological assemblages representing past societies offer information on individual year of birth. Without longitudinal or diachronic analysis of health change, observations of health using bioarchaeological data are limited to a static notion of health surrounding a transition in economy or subsistence. While data on contemporary societies capture gradual changes in health (including emotional and mental health) and quantify stress and specific disease load via bioassays, they lack depth of time available in archaeological assemblages.

Identifying the motivation to transition to a new economic or subsistence regime from archaeological data is challenging but identifying the impact on health is not. We investigate long-term trends in childhood stress using a common bioarchaeological marker of stress, enamel hypoplasias, in the Hadza, a hunter-gatherer population that is slowly but not collectively making the transition to market integration and small-scale agriculture. The bush lifestyle is increasingly at threat due to restrictions on Hadza mobility due to the activities of farmers and pastoralists and the introduction of market economics through ecotourism and researchers paying them for their labor (19).

The Hadza are one of the last hunter-gatherer populations on the planet that still maintain traditional lifestyles. Extant Hadza number approximately 1000 but only roughly 100-300 are still fully engaged in traditional mobile hunter-gatherer subsistence in the bus (20, 21). The ones still practicing hunting-gathering are struggling to survive because access to the land is increasingly restricted by the activities of farmers and pastoralists (19). Bush Hadza live in mobile bush camps of ~30 individuals (22). Camp membership is fluid with people moving in and out of camps frequently (23). Camps change location every six weeks on average but, in recent years, they move less frequently. Bush-living Hadza forage daily for food (22) with foraging practices varying by sex. Hadza men typically forage alone eating berries and baobab fruit while hunting birds and mammals using bows and arrows—the latter may be poisoned arrows for hunting larger game. They also may carry axes to access honey, a preferred food (24). While they eat much of the honey they find, they bring roughly half their haul of medium-to-large game and baobab back to camp (25). Hadza women typically forage in groups of 3-8 adults plus infants, unweaned toddlers, and possible older children. They mainly gather baobab, berries, and several species of tubers using simple, fire-hardened, sharpened branches as digging sticks. Like the men, they eat some of their foraged foods, especially roasted tubers, but take the remainder back to camp along with foraged baobab and berries (22).

In addition to daily food foraging and hunting activities, bush Hadza have been increasingly integrated in market economies from the external forces of research and ethnotourism. Ethnographic research on the Hadza started in 1957. By the 1990s, there were at least 2-3 researchers working in Hadzaland. Researchers have sustained their seasonal presence over the past 30 years and today, there are currently several research groups, each with several senior researchers and several Ph.D. students. This can translate to an average annual presence of 10-15 researchers in Hadzaland, and this can be higher in the summers when the academic calendar allows for more fieldwork. Since the mid-1980s, Hadzaland has been an ecotourism destination (19) and has increased from one bus per day to hundreds of busses per day passing through particular villages. Over 400,000 tourists visit the nearby Ngorongoro Conservation Area each year and the proximity to Hadzaland (a daytrip) increases the attractiveness of ethnotourism in the regions (20). While the increase in researchers has been relatively minimal compared to that of tourists, both have had negative impacts on the Hadza (19) despite any gains from increased flow of goods and medicine and access to schools.

The Hadza are an ideal group to examine because we know their transition is externally motivated and that the transition is happening to both those already one generation into sedentism but also those struggling to maintain the bush lifestyle. Dental enamel hypoplasias are an ideal marker of stress-related disruptions to growth and development (10, 26) because teeth do not regenerate, which means that the horizontal lines appearing in tooth enamel as a result of growth interruptions are a permanent record of childhood health (27). Bioarchaeologists use enamel hypoplasia frequency to assess adult frailty, which is marked

by decreased longevity (28). While sex-based health differences are not commonly found in archaeological assemblages of hunter-gatherers, the few instances in which they have been explained as differential investment in children rather than inherent biological differences in buffering stress (29). Studies of enamel hypoplasia frequency in living populations indicates that higher frequency of defects is commonly observed in those with lower quality of living conditions and lower mean age-at-death (10, 26, 30). Using this bioarchaeological method on living populations provides us a non-invasive measure of childhood health that can be used to determine if men and women have health differences either arising from the subtle differences in status due to a sex-based division of labor or from the transition to a market economy.

METHODS

Ethics: This research was approved by Florida State University IRB (IRB00000446) as well as by the ethics review board of the Tanzania Commission for Science and Technology (COSTECH). Verbal informed consent was obtained, as the majority of Hadza people do not read or write.

Data Collection Methods. Enamel hypoplasia data were collected by JCB on 101 Hadza from six different highly mobile bush camps on individuals who obtained the majority of their daily calories (estimated 90%) from daily food foraging during the data collection period, 2007-2009 (24). Data collection was limited to anterior maxillary teeth (central incisors, lateral incisors, canines) due to the challenges associated with collecting data on posterior dentition in living persons. The presence of enamel hypoplasia on anterior teeth indicates stress occurred during early childhood, or between the ages of 1-5 (30). Teeth were visually inspected for inclusion in the study using an LED light which aided identification and recording of features that obscured crown surface (e.g., chips, stains, worn enamel) and would prevent accurate recording of possible defects.

Hadza women have greater anterior tooth wear because they use their teeth to cut beading string and chew leather hides and bow strings to make them supple (31, 32). Men use their teeth to work arrows, tighten bow strings, and strip bark from branches to make twine (31, 32) and had less wear than women. Differences in diet (33) may also have increased wear in women. Because differences in tooth wear might result in men appearing to have more hypoplasias, any tooth with wear greater than 50% was cut from the analysis (34). Anyone under the age of 18 was also eliminated. Enamel hypoplasia frequency was derived from raw counts per individual by taking the ratio of maximum total number of defects observed to total number of teeth observed.

Data Analysis Methods. Analysis was completed by KCH in R 4.1.2 ‘Bird Hippie’ (35) using R Studio 2021.09.1 Build 372 (36). Tidyverse was used for data cleaning (37). Sample summary and descriptive statistics were generated using psych (38). Enamel hypoplasia frequency data were plotted against the normal distribution using qqplot in car (39). Figures were generated using ggplot2 (40), ggpubr (41), and gridBase (42) using the color-blind friendly palette viridis.

Year of birth and/or age was determined from long-standing demographic data from Nicholas Blurton-Jones and then Frank W. Marlowe, currently curated and maintained by JCB. Birth cohort uses year of birth to determine if an individual had maxillary anterior teeth developing prior to the beginnings of market integration or after market-integration had begun. Market integration of bush Hadza was already in place by the mid-1980s (19, 20)

Since maxillary anterior teeth develop between ages 1-5 (43), stress from market integration would not be reflected in the anterior dentition of individuals born prior to 1975. Thus, we used 1975 as the break point to create a two-factor birth cohort variable, before integration and after integration.

RESULTS

The final sample of usable data (teeth not eliminated due to wear) was 85 (Table 1). Given that only 100-300 adult Hadza continue to engage in traditional hunter-gatherer subsistence economies, our sample size of 85 is between 28% of the total available sample on the low end and 85% on the high end—even the low-end estimate of 28% is a representative sample for hypothesis testing. While we have more women than men, our groups within the sample (birth cohort, sex, sex-based birth cohort) are balanced. The mean age is 37 (median=33) and mean birth year is 1971 (median=1974) with ranges of 62 for both. The mean birth year of those born before 1975 is 1959 (median = 1963) and those born in 1975 or later is 1981 (median = 1981).

Table 1: Sample Demographic

Variables	Group	n	Mean	StErr	StDev	Median	Range
Age	All	85	37.07	1.54	14.20	33.00	62.00
	Men	36	38.39	2.04	12.22	36.50	46.00
	Women	49	36.10	2.22	15.55	33.00	62.00
	1974-	41	48.46	1.89	12.12	45.00	45.00
	1975+	44	26.45	0.65	4.30	26.00	15.00
	Men 1974-	19	48.05	1.96	8.56	47.00	32.00
	Women 1974-	22	48.82	3.14	14.72	44.50	45.00
	Men 1975+	17	27.59	0.75	3.08	29.00	11.00
	Women 1975+	27	25.74	0.93	4.84	25.00	15.00
Birth Cohort	All	85	1.52	0.05	0.50	2.00	1.00
	Men	36	1.47	0.08	0.51	1.00	1.00
	Women	49	1.55	0.07	0.50	2.00	1.00
	1974-	41	1.00	0.00	0.00	1.00	0.00
	1975+	44	2.00	0.00	0.00	2.00	0.00
	Men 1974-	19	1.00	0.00	0.00	1.00	0.00
	Women 1974-	22	1.00	0.00	0.00	1.00	0.00
	Men 1975+	17	2.00	0.00	0.00	2.00	0.00
	Women 1975+	27	2.00	0.00	0.00	2.00	0.00
Birth Year	All	85	1971	1.52	13.99	1974	62.00
	Men	36	1969	2.04	12.22	1971	45.00
	Women	49	1971	2.17	15.22	1975	62.00
	1974-	41	1959	1.87	11.98	1963	44.00
	1975+	44	1981	0.63	4.19	1981	17.00
	Men 1974-	19	1960	2.00	8.70	1960	32.00
	Women 1974-	22	1959	3.08	14.43	1964	43.00
	Men 1975+	17	1980	0.76	3.13	1979	11.00
	Women 1975+	27	1982	0.90	4.70	1982	17.00

An average of eight teeth were observed in men and women both. Women had more teeth (a median of 11) compared to men (a median of 6.5) and a larger hypoplasia count. When considering the frequency (to eliminate the effect of difference in number of teeth observed), women had a greater mean and median frequency of hypoplasia (count of hypoplasias

divided by total number of teeth observed) (Table 2). Average hypoplasia frequency drops by 40% in the birth cohort 1975+ but this is driven by change in men's hypoplasia frequency which drops by 67% after market integration (compared to the 20% drop in women).

Table 2: Descriptive Statistics

Variables	Group	n	Mean	StErr	StDev	Median	Range
Hypo Frequency	All	81	0.80	0.08	0.71	0.67	2.50
	Men	33	0.75	0.13	0.74	0.55	2.50
	Women	48	0.84	0.10	0.69	0.78	2.00
	1974-	39	1.03	0.12	0.74	1.00	2.50
	1975+	42	0.59	0.10	0.62	0.50	2.00
	Men 1974-	17	1.13	0.19	0.80	1.00	2.50
	Women 1974-	22	0.94	0.15	0.70	0.92	2.00
	Men 1975+	16	0.34	0.10	0.38	0.17	1.00
	Women 1975+	26	0.74	0.13	0.69	0.58	2.00
Total Hypos	All	85	5.61	0.53	4.87	4.00	16.00
	Men	36	4.53	0.71	4.29	4.00	14.00
	Women	49	6.41	0.74	5.16	6.00	16.00
	1974-	41	6.37	0.75	4.81	6.00	16.00
	1975+	44	4.91	0.73	4.87	4.00	16.00
	Men 1974-	19	5.74	1.00	4.37	4.00	14.00
	Women 1974-	22	6.91	1.11	5.20	6.50	16.00
	Men 1975+	17	3.18	0.94	3.88	2.00	12.00
	Women 1975+	27	6.00	1.00	5.18	6.00	16.00
Total Teeth	All	85	8.27	0.41	3.76	9.00	12.00
	Men	36	7.75	0.66	3.94	6.50	12.00
	Women	49	8.65	0.52	3.61	11.00	12.00
	1974-	41	7.12	0.57	3.67	6.00	12.00
	1975+	44	9.34	0.54	3.56	12.00	12.00
	Men 1974-	19	5.95	0.78	3.39	6.00	12.00
	Women 1974-	22	8.14	0.78	3.67	6.00	11.00
	Men 1975+	17	9.76	0.87	3.58	12.00	12.00
	Women 1975+	27	9.07	0.69	3.58	12.00	12.00

The left tail of hypoplasia frequency data distribution skews the data outside the range of the normal distribution (Figure 1). The generalized linear model (GLM) is tolerant to deviations from the normal distribution and was used for regression models (44).

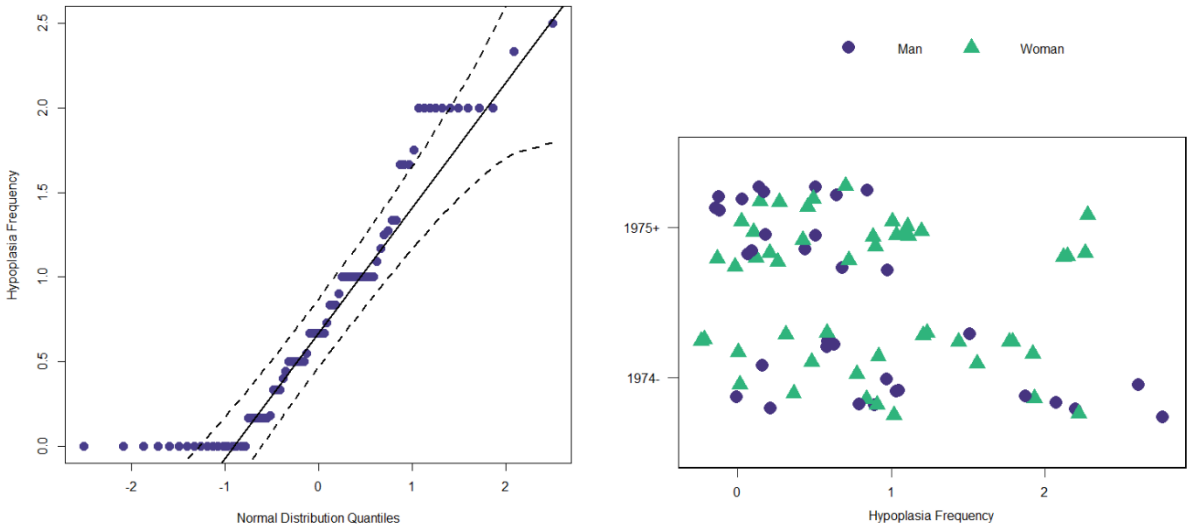

Figure 1: Hypoplasia Frequency Data Distribution. Against the Normal Curve (Left) and By Sex and Birth Cohort (Right).

When examined from bioarchaeological perspective, there are no significant differences between men and women even if women have slightly higher hypoplasia frequencies. When examined from the perspective of living populations and the benefit of year of birth, we can see distinct temporal trends across the dataset where men experience a significant decrease in hypoplasia frequency and there is not significant change in women—these values were noted previously in the descriptive data where the men experienced a trebling of the men and women experienced only minor change. The slope of change over time indicates that change in women is not significant statistically. We also see that around the time of market integration in the 1970s, men decrease from having at least 1 hypoplasia to having almost none (on average across men).

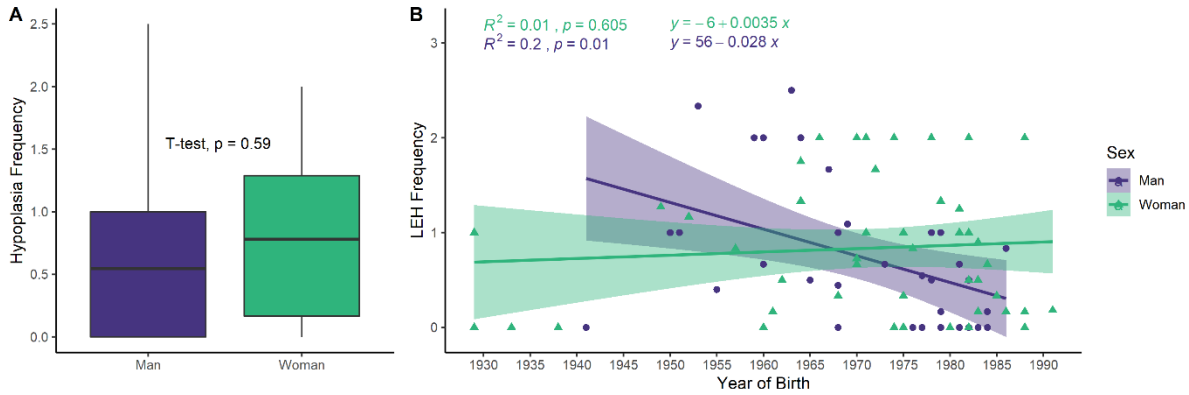

Figure 2: LEH frequency relative to year of birth, by sex

CONCLUSIONS AND IMPLICATIONS

A previous study compared enamel hypoplasia frequencies between 17 bush-living Hadza and 34 village-living Hadza (45) and found that men, but not women, had higher enamel hypoplasia frequencies in village settings than bush settings. Life is worse for men in the villages than the bush and may be due to differences in weaning (45), but also there may be changes in diet (14), sanitation (15), family structure (11, 16, 17), patterns of parental investment, to name a few factors. Indeed, hunter-gatherers spending more time in sedentary

villages have higher parasite and viral loads than those in the same ethnolinguistic group still foraging (18). With a larger sample and factoring by birth cohort as a proxy for market integration, our results indicate that life in the bush is not better for everyone. Life in the bush is better for men as external market forces are introduced and adopted.

Why do bush-living Hadza women sustain stress after they have been integrated into a market economy? Explanations for sex-based health disparity include changes in wealth, school attendance, parental provisioning, and medical care following the incorporation of Hadzaland into the Tanzanian state. The most compelling explanation is that of wealth disparity arising from increased presence of researchers and tourists. About 29% of Hadza research focuses primarily on one sex and men are featured seven times than women (Appendix 1). Researcher presence may have less impact in terms of sheer numbers when compared to the increasing numbers of ethnotourists starting in the mid-1980s. Wealth disparity has been a documented side effect of ethnotourism in other egalitarian groups (46-48). Among the Hadza, men are paid regularly to take tourists on a hunt. Because the activities of Hadza men are externally valued more than women, men enter the market economy as wage laborers and exchange their wages for valuable market goods produced outside the community (e.g., cigarettes, alcohol, but also grains and other agricultural products). Women have some access to valuable market goods, but they have less control over the distribution of goods than men. The introduction of market forces changed other social factors like medical care and education. Ability to provide medical care to a young child by paying for transport and medicine may be a significant factor—especially when coupled with differential positive provisioning for boys by fathers/parents and lack of provisioning for females—a model seen in Tanzania already (49) but not yet studied in Hadza. Hadza parents may be preferentially provisioning male children with certain foods or possibly weaning them differently, as has been observed in nearby communities (49). Weaning differences have not been investigated yet in the Hadza. Education allows access to calories from school means and Hadza boys are preferentially sent school over girls (21: page 36). The timing of the defects we studied on the upper anterior dentition indicate very early childhood stress—before 5 years of age—which suggests education may increase disparity with age but is not the root source.

In the absence of a long-term ethnographic study on health, there are limited options for understanding how population health changes over time in small-scale societies. We applied a method of assessing general childhood stress commonly used in bioarchaeology to a living human population, the highly mobile, bush-living Hadza hunter-gatherers—a group for which we have decades of ethnographic data. The method we used to examine enamel hypoplasias does not require expensive equipment, is non-invasive, and allows analysis of long-term changes in health which gives researchers a glimpse into a population's health conditions 30, or even 60 years ago. Comparative data could be easily collected on indigenous populations having undergone market integration in the past 2-3 generations. Even if integration is now complete, there will be elderly adults with a record of health disruptions during their childhood before, during, and after the major cultural change. This will then allow us to re-examine inferences about transitions in past societies and identify new techniques to test whether we are missing data. An example of a possible path forward is a promising new technique for extracting stress hormones from bone (50) but has yet to be applied to human remains. If it is possible to extract stress from ancient bones, we would only need to find a way to compare those results to stress hormones extracted from living tissue. By applying bioarchaeological techniques to living populations, we can gain

longitudinal insights into societal health and directly observe what has been inferred from archaeological assemblages which have the advantage of time depth.

Market integration has been understood as an increase in female morbidity, but we find the opposite—a decrease in male morbidity and no change to female health. Further, we associated the change in male childhood stress (and lack of change in female childhood stress) with the timing of the introduction of market forces. If the health disparity remains the same, why is distinguishing between benefit versus disadvantage important? Our data indicate that gender inequality and wealth disparity is an early symptom of transition from egalitarianism (often hunter-gatherers) to hierarchical systems (often food producing). The differences between these two paths (increased health benefits going to males versus increased morbidity in females) would not be visible in a bioarchaeological context due to the lack of dating accuracy in burial deposition and have not been witnessed in living transitional populations due to the lack of longitudinal health data.

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References

1. Mattison SM, Smith EA, Shenk MK, et al.; The evolution of inequality. *Evolutionary Anthropology: Issues, News, and Reviews* 2016;**25**(4):184-199.
2. Price TD, Feinman GM; Foundations of Social Inequality. New York: Plenum Press, 1995.
3. Marlowe FW; Hunting and gathering: The human sexual division of foraging labor. *Cross-Cultural Research* 2007;**41**(2):170–95.
4. Fedurek P, Lacroix L, Lehmann J, et al.; Status does not predict stress: Women in an egalitarian hunter–gatherer society. *Evolutionary Human Sciences* 2020;**2**:e44. doi: 10.1017/ehs.2020.44.
5. Kent S; Does sedentarization promote gender inequality? A case study from the Kalahari. *The Journal of the Royal Anthropological Institute* 1995;**1**(3):513-536.
6. Dyble M, Salali GD, Chaudhary N, et al.; Sex equality can explain the unique social structure of hunter-gatherer bands. *Science* 2015;**348**(6236):796-798. doi: 10.1126/science.aaa5139.
7. Armelagos GJ, Swedlund AC; Disease in Populations in Transition: Anthropological and Epidemiological Perspectives Santa Barbara: Praeger, 1990.
8. Goodman AH; The chronology of enamel hypoplasias in an industrial population: A reappraisal of Sarnat and Shour (1941, 1942). *Human Biology* 1988;**60**(5):781-791.
9. Goodman AH, Leatherman TL; Building a New Biocultural Synthesis: Political-Economic Perspectives on Human Biology. Ann Arbor: University of Michigan Press, 1998.
10. Goodman AH, Thomas RB, Swedlund AC, et al.; Biocultural perspectives on stress in prehistoric, historical, and contemporary population research. *American Journal of Physical Anthropology* 1988;**31**(S9):169-202.
11. Kramer KL, Greaves RD; Changing Patterns of Infant Mortality and Maternal Fertility among Pumé Foragers and Horticulturalists. *American Anthropologist* 2007;**109**(4):713-726. doi: 10.1525/aa.2007.109.4.713.

12. Page AE, Minter T, Viguier S, et al.; Hunter-gatherer health and development policy: How the promotion of sedentism worsens the Agta's health outcomes. *Social Science & Medicine* 2018;**197**:39-48. doi: 10.1016/j.socscimed.2017.12.002.
13. Lea AJ, Waigwa C, Muhoya B, et al.; Socioeconomic status effects on health vary between rural and urban Turkana. *Evolution, Medicine, and Public Health* 2021:eoab039. doi: 10.1093/emph/eoab039.
14. Dounias E, Froment A; From Foraging to Farming Among Present-Day Forest Hunter-Gatherers: Consequences on Diet and Health. *International Forestry Review* 2011;**13**(3):294-304, 11.
15. Caruso BA, Sevilimedu V, Fung ICH, et al.; Gender disparities in water, sanitation, and global health. *The Lancet* 2015;**386**(9994):650-651.
16. Khera R, Jain S, Lodha R, et al.; Gender bias in child care and child health: global patterns. *Archives of disease in childhood* 2014;**99**(4):369-374.
17. Osmani S, Sen A; The hidden penalties of gender inequality: fetal origins of ill-health. *Economics & Human Biology* 2003;**1**(1):105-121.
18. Page AE, Viguier S, Dyble M, et al.; Reproductive trade-offs in extant hunter-gatherers suggest adaptive mechanism for the Neolithic expansion. *Proceedings of the National Academy of Sciences of the United States of America* 2016;**113**(17):4694-4699. doi: 10.1073/pnas.1524031113.
19. Gibbons A; Hadza on the brink. *Science* 2018;**360**(6390):700-704. doi: 10.1126/science.360.6390.700.
20. Yatsuka H; Reconsidering the "indigenous peoples" in the African context from the perspective of current livelihood and its historical changes: The case of the Sandawe and the Hadza in Tanzania. *African Study Monographs* 2015;**36**(1):27-48.
21. Marlowe F. *The Hadza: Hunter-Gatherers of Tanzania.* Berkeley: University of California Press, 2010.
22. Marlowe FW; Central place provisioning: The Hadza as an example. In: Hohmann G, Robbins M, C. Bs (eds). *Feeding ecology in apes and other primates* Cambridge: Cambridge University Press, 2006, 357-375.
23. Blurton Jones N, Hawkes K, O'Connell JF; Older Hadza men and women as helpers: Residence data. In: Hewlett BS, Lamb MEs (eds). *Hunter-Gatherer Childhoods: Evolutionary, developmental, and Cultural Perspectives*. New Brunswick: Transaction Publishers, 2005, 214-236.
24. Marlowe FW, Berbesque JC, Wood B, et al.; Honey, Hadza, hunter-gatherers, and human evolution. *Journal of Human Evolution* 2014;**71**:119-128. doi: https://doi.org/10.1016/j.jhevol.2014.03.006.
25. Berbesque JC, Wood BM, Crittenden AN, et al.; Eat first, share later: Hadza hunter–gatherer men consume more while foraging than in central places. *Evolution and Human Behavior* 2016;**37**(4):281-286. doi: https://doi.org/10.1016/j.evolhumbehav.2016.01.003.
26. Goodman AH, Rose JC; Assessment of systemic physiological perturbations from dental enamel hypoplasias and associated histological structures. *Yearbook of Physical Anthropology* 1990;**33**(S11):59-110. doi: 10.1002/ajpa.1330330506.
27. Goodman AH, Martin DL; Reconstructing health profiles from skeletal remains. In: Steckel RH, Rose JCs (eds). *The Backbone of History: Health and Nutrition in the Western Hemisphere*. New York: Cambridge University Press, 2002, 11-60.
28. Yaussy SL, DeWitte SN, Redfern RC; Frailty and famine: Patterns of mortality and physiological stress among victims of famine in medieval London. *American Journal of Physical Anthropology* 2016;**160**(2):272-83. doi: 10.1002/ajpa.22954.

29. Guatelli-Steinberg D, Lukacs JR; Interpreting sex differences in enamel hypoplasia in human and non-human primates: Developmental, environmental, and cultural considerations. *American Journal of Physical Anthropology* 1999;**110**(29):73-126.
30. Goodman AH, Armelagos GJ; Factors affecting the distribution of enamel hypoplasias within the human permanent dentition. *American Journal of Physical Anthropology* 1985;**68**:479-493.
31. Berbesque JC, Marlowe FW, Pawn I, et al.; Sex differences in Hadza hunter-gatherer dental wear: A preliminary report. . *Human Nature* 2012; **23**:270-282. doi: 10.1007/s12110-012-9145-9.
32. Berbesque JC, Marlowe FW, Pawn I, et al.; Sex Differences in Hadza Dental Wear Patterns. *Human Nature* 2012;**23**(3):270-282. doi: 10.1007/s12110-012-9145-9.
33. Berbesque JC, Marlowe FW, Crittenden AN; Sex differences in Hadza eating frequency by food type. *Am J Hum Biol* 2011;**23**(3):339-45. doi: 10.1002/ajhb.21139.
34. Reid DJ, Dean MC; Variation in modern human enamel formation times. *Journal of Human Evolution* 2006;**50**(3):329-346. doi: https://doi.org/10.1016/j.jhevol.2005.09.003.
35. R Development Core Team; R: A language and environment for statistical computing. Vienna, Austria, 2018.
36. RStudio Team; RStudio: Integrated Development for R. Boston, MA: RStudio, Inc., 2020.
37. Wickham H, Averick M, Bryan J, et al.; Welcome to the tidyverse. *Journal of Open Source Software* 2019;**4**(43):1686. doi: https://doi.org/10.21105/joss.01686.
38. Revelle W; psych: Procedures for Personality and Psychological Research, Northwestern University, Evanston, Illinois, USA, https://CRAN.R-project.org/package=psych Version = 1.8.12., 2018.
39. Fox J, Weisberg S. *An {R} Companion to Applied Regression.* Thousand Oaks CA: Sage, 2011.
40. Wickham H. *ggplot2: Elegant Graphics for Data Analysis.* New York: Springer-Verlag, 2016.
41. Kassambara A; ggpubr: 'ggplot2' Based Publication Ready Plots. R package version 0.2. https://CRAN.R-project.org/package=ggpubr. 2018.
42. Murrell P; gridBase: Integration of base and grid graphics R package version 0.4-7 https://CRAN.R-project.org/package=gridBas. 2014.
43. AlQahtani SJ, Hector MP, Liversidge HM; Accuracy of dental age estimation charts: Schour and Massler, Ubelaker and the London Atlas. *American Journal of Physical Anthropology* 2014;**154**(1):70-8. doi: 10.1002/ajpa.22473.
44. Fox J. *Applied Regression Analysis and Generalized Linear Models.* Thousand Oaks, CA: Sage, 2008.
45. Ungar PS, Crittenden AN, Rose JC; Toddlers in transition: linear enamel hypoplasias in the Hadza of Tanzania. *International Journal of Osteoarchaeology* 2017;**27**(4):638-649. doi: 10.1002/oa.2586.
46. Wierucka A; Living with strangers: Huaorani and tourism industry in the 21 st century. *Anthropological Notebooks* 2018;**24**(1).
47. Guenther M; Contemporary Bushman Trance Dancer and Trance Dance, and the Decline of Sharing. In: Widlok T, Tadess WGs (eds). *Property and Equality: Encapsulation, commercialisation, discrimination*, 2005, 208-230.
48. Gezon LL; Who wins and who loses? Unpacking the “local people” concept in ecotourism: a longitudinal study of community equity in Ankarana, Madagascar. *Journal of Sustainable Tourism* 2014;**22**(5):821-838.

49. Hassan A, Schaffnit SB, Sear R, et al.; Fathers favour sons, mothers don't discriminate: Sex-biased parental care in northwestern Tanzania. *Evolutionary Human Sciences* 2019;**1**:e13. doi: 10.1017/ehs.2019.14.
50. Charapata P, Horstmann L, Jannasch A, et al.; A novel method to measure steroid hormone concentrations in walrus bone from archaeological, historical, and modern time periods using liquid chromatography/tandem mass spectrometry. *Rapid Communications in Mass Spectrometry* 2018;**32**(23):1999-2023. doi: 10.1002/rcm.8272.